

Implementation Cheat Sheet: Lengnick (2013) Macroeconomic Model

Agent-Based Macroeconomics: Baseline Specification, Entity Schemas & Process Dynamics

Authoritative reference: Matthias Lengnick (2013), *Journal of Economic Behavior & Organization*, Vol. 86, pp. 102–120.

1. Executive Summary & Architectural Overview

Decentralized Macroeconomic Microfoundations

The Lengnick model is a benchmark decentralized, out-of-equilibrium agent-based macroeconomic model featuring **two agent classes** (Households and Firms) without a central Walrasian auctioneer, commercial banks, or government interventions.

PHASE 1: MONTHLY PLANNING

Day 1 Morning (Every 21 Ticks):

- Firms review wages (w_f^{new})
- Labor buffer targets (i_f vs $\phi, \bar{\phi}$)
- Calvo sticky price adjustments (p_f)
- Type A supplier rewiring ($n = 7$)
- Labor market matching (w_f vs ω_h)
- Keynesian consumption budget (c_h^r)

PHASE 2: DAILY OPERATIONS

Days 1 to 21 (Daily Trading Loop):

- Randomized household shopping
- Sequential visits across 7 suppliers
- Cash-in-advance purchases (q^*)
- Firm daily linear production ($y_f = \lambda l_f$)
- Inventory accumulation ($i_f + = y_f$)
- Unmet demand logged for rewiring

PHASE 3: MONTH-END SETTLEMENT

Day 21 Close (Accounting Reset):

- Contractual wage payments ($W_f = w_f l_f$)
- Aggregate profit clearing (Π_{total})
- Wealth-proportional dividends
- Firm cash reset ($m_f = 0$ strict)
- Reservation wage adaptation (ω_h)
- Sever deferred firings & roll time

Key Architectural Tenets:

- Dual Time Horizons:** Daily operational frequency ($t = 1, \dots, T_{\text{sim}} \cdot 21$) for goods transactions and physical production; monthly accounting frequency ($T = 21$ business days) for labor contracts, price revisions, and wage disbursements.
- Decentralized Bipartite Networks:**
 - Type A (Commercial Links):** Directed buyer-seller connections. Each household is linked to exactly $n = 7$ supplier firms (out-degree = 7). Households possess zero global price information.
 - Type B (Employment Links):** Directed employer-worker contracts. Each worker has at most 1 employer (in-degree $\in \{0, 1\}$); each firm employs l_f workers (out-degree = l_f).
- Pure Exchange Economy & Strict Money Conservation:** Money functions purely as a medium of exchange. Total money stock M_{total} is strictly invariant across all daily transactions and monthly settlements:

$$\sum_{h=1}^H m_h + \sum_{f=1}^F m_f = M_{\text{total}} \quad (\forall t)$$

At the close of each month, all firm cash balances are fully distributed, enforcing $m_f = 0$ for all firms at the start of every month.

2. Global Configuration & Parameter Reference

Table 1 (Lengnick 2013, p. 105)

The baseline model is parameterized to calibrate an economy of 1000 workers and 100 firms over a 500-year horizon ($T_{\text{sim}} = 6000$ months) preceded by a 1000-month initialization burn-in:

Parameter Name	Symbol	Scope	Value	Description & Behavioral Function
Households Count	H	Global	1000	Total workforce / household agents. Inelastic labor supply of 1 unit each.
Firms Count	F	Global	100	Total enterprise agents (fixed size; no firm entry/exit in baseline model).
Days per Month	T	Global	21	Business trading days per macro planning month (1 year = $12 \cdot 21 = 252$ days).
Consumption Exponent	α	Household	0.9	Non-linear Keynesian marginal propensity to consume curvature ($0 < \alpha < 1$).
Supplier Network Size	n	Household	7	Exact out-degree of commercial Type A links per household.
Price Search Prob.	ψ_{Price}	Household	0.25	Monthly probability a household seeks a cheaper supplier among larger firms.
Price Discount Threshold	ξ	Household	0.01	Minimum price reduction required (1%) to trigger replacing a commercial link.
Stockout Rewire Prob.	ψ_{Quant}	Household	0.25	Monthly probability a quantity-constrained buyer replaces a stockout supplier.
Unemployed Search Scope	β	Household	5	Maximum number of random firms visited per month by an unemployed jobseeker.
On-the-Job Search Prob.	π	Household	0.1	Monthly probability that an employed, satisfied worker initiates on-the-job search.
Reservation Markdown	$\Delta\omega$	Household	0.10	Fractional reduction (10%) applied to reservation wage ω_h after an idle month.
Labor Productivity	λ	Firm	3	Daily goods produced per employed worker: $y_f = \lambda \cdot l_f = 3 \cdot l_f$.
Wage Shock Bound	δ	Firm	0.019	Upper bound of uniform wage adjustment shock: $\mu \sim U[0, \delta]$.

Parameter Name	Symbol	Scope	Value	Description & Behavioral Function
Full Employment Memory	γ	Firm	24	Consecutive months operating at full staffing required before lowering wage rate.
Inventory Lower Bound	$\underline{\phi}$	Firm	0.25	Lower inventory buffer: $\underline{i}_f = \underline{\phi} \cdot d_f^{\text{old}}$ (triggers hiring / price hikes).
Inventory Upper Bound	$\bar{\phi}$	Firm	1.0	Upper inventory buffer: $\bar{i}_f = \bar{\phi} \cdot d_f^{\text{old}}$ (triggers firings / price cuts).
Price Markup Min Bound	$\underline{\varphi}$	Firm	1.025	Lower bound on goods price relative to marginal cost: $\underline{p}_f = \underline{\varphi} \cdot mc_f$.
Price Markup Max Bound	$\bar{\varphi}$	Firm	1.15	Upper bound on goods price relative to marginal cost: $\bar{p}_f = \bar{\varphi} \cdot mc_f$.
Price Adjustment Bound	ϑ	Firm	0.02	Upper bound of uniform price adjustment step: $\nu \sim U[0, \vartheta]$.
Calvo Sticky Probability	θ	Firm	0.75	Calvo price friction: probability of successfully implementing a desired price change.
Simulation Horizon	T_{sim}	Global	6000	500 years (6000 months), run after discarding a 1000-month initialization transient.

3. Entity Tables & Schema Specifications

Data Types, Invariants & Foreign Keys

3.1 Household Entity Schema (H = 1000)

Each household inelastically supplies 1 unit of labor, receives income (wages and profit shares), and purchases consumption goods:

Field Name	Data Type	Constraint / Default	Role & Behavioral Meaning
id	int	Primary Key: 1...H	Unique household identifier.
liquidity (m_h)	float	$m_h \geq 0$	Cash holdings. Bounded by cash-in-advance; cannot be over-drawn.
reservation_wage (ω_h)	float	$\omega_h > 0$	Minimum acceptable wage for employment. Ratchets on wage hike; drops 10% when idle.
employer_id	int null	Foreign Key $\rightarrow$ Firm	Type B connection. Current employer (null if worker is currently unemployed).
supplier_ids	List[int]	Size supplier_ids = 7	Type A connections to 7 supplier firms. Directed buyer-seller network.
monthly_real_budget (c_h^r)	float	$c_h^r \geq 0$	Planned real consumption volume for current month: $\min\left\{\left(\frac{m_h}{p_h^I}\right)^\alpha, \frac{m_h}{p_h^I}\right\}$.
daily_real_demand (c_h^{daily})	float	$= \frac{c_h^r}{21}$	Daily consumption demand quota target evaluated each morning.
constrained_firms	Dict[int, float]	Default: {}	Maps supplier firm ID to unsatisfied real demand volume caused by inventory stockouts.
is_employed	bool	Derived (employer != null)	Convenience status flag indicating employment status (in-degree == 1).
is_satisfied	bool	Derived ($w_{\text{empl}} \geq \omega_h$)	Job satisfaction indicator. Satisfied workers search with $\pi = 0.1$; unsatisfied search with 1.0.

3.2 Firm Entity Schema (F = 100)

Firms employ workers, produce consumption goods with linear technology, manage inventories, set prices and wages, and disburse profits:

Field Name	Data Type	Constraint / Default	Role & Behavioral Meaning
id	int	Primary Key: 1...F	Unique enterprise identifier.
liquidity (m_f)	float	Signed float	Operational cash balance. Accumulates sales revenue; deducted for payroll; cleared to 0 monthly.
inventory (i_f)	float	$i_f \geq 0$	Stock of finished goods ready for customer dispatch. Increased by production; reduced by sales.
price (p_f)	float	$p_f \in [\underline{p}_f, \bar{p}_f]$	Current selling price per unit. Sticky adjustment via Calvo parameter $\theta = 0.75$.
wage (w_f)	float	$w_f > 0$	Monthly wage offered to all existing workers and new recruits.
employees	Set[int]	Foreign Keys $\rightarrow$ Household	Set of active employee IDs (size $l_f = \text{employees} $). Determines daily production batch.
vacancies	int	$\in \{0, 1\}$	Number of open positions offered this month. Set to 1 if $i_f < \underline{i}_f$, else 0.
unfilled_vacancy_last_month	bool	Default: false	Flag indicating whether firm posted a vacancy last month that failed to attract a hire.
consecutive_full_months	int	≥ 0	Consecutive months with all positions filled. Triggers wage reduction when ≥ 24 .

Field Name	Data Type	Constraint / Default	Role & Behavioral Meaning
pending_firings	List[int]	Default: []	Employees marked for dismissal at month end due to job protection statutory lag.
demand_last_month (d_f^{old})	float	$d_f^{\text{old}} \geq 0$	Total units demanded in previous month. Anchors inventory buffer thresholds $\underline{i}_f$ and $\bar{i}_f$.
demand_current_month	float	Accumulator (≥ 0)	Running total of real demand intentions received across all 21 trading days.
marginal_cost (mc_f)	float	Derived: $\frac{w_f}{21 \cdot \lambda}$	Unit labor cost of production: $mc_f = \frac{w_f}{63}$. Bounds pricing boundaries.

3.3 Network Topology (Relational Bipartite View)

The macroeconomy is governed by two directed bipartite graphs linking Households and Firms:

- **Type A Links (Goods Market Network):** Directed connection from Household h to Firm f . Out-degree of every household is strictly fixed at $n = 7$. In-degree of firm f is unconstrained ($\in [0, H]$). Captures brand loyalty, geographic separation, and localized information.
- **Type B Links (Labor Market Network):** Directed contract from Firm f to Household h . In-degree of household is at most 1 (0 if unemployed). Out-degree of firm is its workforce size l_f .

4. Process Dynamics: Step-by-Step Algorithmic Logic

Sequential Execution Specifications

⚡ Execution Ordering Policy

The simulation loop must execute in strict chronological sequence. Random agent orderings must be freshly reshuffled before each loop pass to eliminate structural primacy bias.

Phase 1: Beginning-of-Month Operations (Planning & Network Matching)

Executed on the morning of Day 1 ($t \equiv 1 \bmod 21$) before any market transactions.

Step 1.1: Firm Wage Review & Adaptation (PROC-1.1) Each firm f evaluates its past staffing conditions:

- **Labor Shortage (Wage Hike):** If firm had an unfilled vacancy last month (`unfilled_vacancy_last_month == true`):

$$w_f^{\text{new}} = w_f^{\text{old}} \cdot (1 + \mu), \quad \mu \sim U[0, \delta] \quad (\delta = 0.019)$$

Reset counter: `consecutive_full_months = 0`.

- **Labor Surplus (Wage Cut):** Else if firm remained fully staffed for at least $\gamma = 24$ consecutive months (`consecutive_full_months >= 24`):

$$w_f^{\text{new}} = w_f^{\text{old}} \cdot (1 - \mu), \quad \mu \sim U[0, \delta]$$

- **Neutral:** Otherwise, preserve existing wage rate: $w_f^{\text{new}} = w_f^{\text{old}}$.

Step 1.2: Firm Production Planning & Labor Targets (PROC-1.2) Firms evaluate inventory i_f against buffer thresholds derived from previous month demand d_f^{old} :

$$\underline{i}_f = \underline{\phi} \cdot d_f^{\text{old}} = 0.25 \cdot d_f^{\text{old}}, \quad \bar{i}_f = \bar{\phi} \cdot d_f^{\text{old}} = 1.0 \cdot d_f^{\text{old}}$$

- **Case 1 (Inventory Deficit, $i_f < \underline{i}_f$):** Open 1 vacancy: `vacancies = 1`.
- **Case 2 (Inventory Excess, $i_f > \bar{i}_f$ and $l_f > 0$):** Select random worker $h \in \text{employees}$; schedule in `pending_firings` (dismissal executes after 1-month statutory job-protection lag).
- **Case 3 (Normal Buffer, $\underline{i}_f \leq i_f \leq \bar{i}_f$):** No hiring or firing; `vacancies = 0`.

Step 1.3: Firm Calvo Sticky Price Setting (PROC-1.3) Prices adjust only when inventory breaches buffer thresholds:

- Nominal marginal cost: $mc_f = \frac{w_f}{21 \cdot \lambda} = \frac{w_f}{63}$.
- Markup bounds: $\underline{p}_f = \underline{\varphi} \cdot mc_f = 1.025 \cdot mc_f$, and $\bar{p}_f = \bar{\varphi} \cdot mc_f = 1.15 \cdot mc_f$.
- **Price Hike ($i_f < \underline{i}_f$):** If $p_f < \bar{p}_f$, with probability $\theta = 0.75$: draw $\nu \sim U[0, \vartheta]$ ($\vartheta = 0.02$) and set $p_f^{\text{new}} = p_f^{\text{old}} \cdot (1 + \nu)$.
- **Price Cut ($i_f > \bar{i}_f$):** If $p_f > \underline{p}_f$, with probability $\theta = 0.75$: draw $\nu \sim U[0, \vartheta]$ and set $p_f^{\text{new}} = p_f^{\text{old}} \cdot (1 - \nu)$.
- **Sticky Default:** With probability $1 - \theta = 0.25$, or inside buffer, price remains unchanged.

Step 1.4: Consumer Network Rewiring (PROC-1.4) Households adapt their 7-supplier commercial network:

1. **Price Search (Prob $\psi_{\text{Price}} = 0.25$):**
 - Pick connected firm $f_{\text{curr}} \in \text{supplier_ids}_h$ uniformly at random.
 - Sample candidate firm $f_{\text{cand}} \notin \text{supplier_ids}_h$ with probability proportional to workforce size: $P(f_{\text{cand}} = k) \propto l_k$.
 - If $p_{f_{\text{cand}}} \leq (1 - \xi) \cdot p_{f_{\text{curr}}}$ (at least $\xi = 1\%$ cheaper): swap link (remove f_{curr} , insert f_{cand}).
2. **Quantity-Constraint Search (Prob $\psi_{\text{Quant}} = 0.25$):**
 - If `constrained_firms` is non-empty: select f_{bad} with probability proportional to unmet demand magnitude.
 - Pick unconnected firm f_{new} uniformly at random; swap link (remove f_{bad} , insert f_{new}).

Step 1.5: Labor Market Search & Matching (PROC-1.5) Process households in randomized sequence across three search tiers:

- **Tier 1 (Unemployed Workers):** Visit up to $\beta = 5$ random firms sequentially. If visited firm has `vacancies > 0` and $w_f \geq \omega_h$: worker hired! Form Type B link, decrement f 's vacancies by 1, and terminate search.
- **Tier 2 (Employed & Unsatisfied, $w_{\text{current}} < \omega_h$):** Search with probability 1.0. Visit 1 random firm f . If f has `vacancies > 0` and $w_f > w_{\text{current}}$: switch employer (sever old Type B link, form new Type B link, decrement f 's vacancies by 1).

- **Tier 3 (Employed & Satisfied, $w_{\text{current}} \geq \omega_h$):** Search with probability $\pi = 0.1$. Visit 1 random firm f . If f has vacancies > 0 and $w_f > w_{\text{current}}$: switch employer.

Step 1.6: Household Consumption Budget Planning (PROC-1.6) Households calculate local price average across their 7 suppliers:

$$P_h^I = \frac{1}{7} \sum_{f \in \text{supplier_ids}_h} p_f$$

Planned monthly real consumption follows Keynesian concave demand bounded by cash wealth:

$$c_h^r = \min \left\{ \left(\frac{m_h}{P_h^I} \right)^\alpha, \frac{m_h}{P_h^I} \right\}, \quad \alpha = 0.9$$

Daily real consumption quota: $c_h^{\text{daily}} = \frac{c_h^r}{21}$.

Phase 2: Daily Operations Loop (Days 1 to 21)

Executed every business day $d \in \{1, \dots, 21\}$ in strict sequence: Consumption Market $\rightarrow$ Production.

Step 2.1: Consumption Market Transactions (PROC-2.1)

1. Reshuffle all households into a randomized sequence.
2. For each household h :
 - Set remaining quota: $q_{\text{rem}} = c_h^{\text{daily}}$; visited counter: $k = 0$.
 - While $k < n$ ($k < 7$) and $q_{\text{rem}} > 0.05 \cdot c_h^{\text{daily}}$ (until $\geq 95\%$ fulfilled):
 - Pick unvisited supplier $f \in \text{supplier_ids}_h$; increment $k \rightarrow k + 1$.
 - Max affordable units: $q_{\text{aff}} = \frac{m_h}{p_f}$; Available units: $q_{\text{avail}} = i_f$.
 - Desired units: $q_{\text{des}} = \min(q_{\text{rem}}, q_{\text{aff}})$.
 - Executed trade quantity:

$$q^* = \min(q_{\text{des}}, q_{\text{avail}})$$

- Log demand: $f.\text{demand_current_month} += q_{\text{des}}$.
- Log stockout: If $q^* < q_{\text{des}}$, record unmet ($q_{\text{des}} - q^*$) in $h.\text{constrained_firms}[f]$.
- Settle trade: $m_h \rightarrow m_h - q^* p_f$, $m_f \rightarrow m_f + q^* p_f$, $i_f \rightarrow i_f - q^*$, $q_{\text{rem}} \rightarrow q_{\text{rem}} - q^*$.
- **Any unmet demand at day end vanishes without backlog.**

Step 2.2: Daily Production Process (PROC-2.2)

For every firm $f \in \{1, \dots, F\}$ on day d :

- Daily manufactured output: $y_f = \lambda \cdot l_f = 3 \cdot l_f$.
- Inventory accumulation: $i_f \rightarrow i_f + y_f$.
- Daily Aggregate GDP: $Y_d = \sum_{f=1}^F y_f$.

Phase 3: Month-End Settlement & Revision (Day 21 Night)

Executed at the conclusion of day 21. Handles accounting settlements, cash reset, and reservation wages.

Step 3.1: Payroll & Wage Disbursement (PROC-3.1)

For each firm f :

- Total payroll obligation: $W_f = w_f \cdot l_f$.
- Deduct wage bill: $m_f \rightarrow m_f - W_f$ (**Note: m_f may turn negative if monthly sales revenue was below payroll**).
- For each employee $h \in \text{employees}_f$: transfer wage: $m_h \rightarrow m_h + w_f$.

Step 3.2: Profit Redistribution & Strict Firm Cash Reset (PROC-3.2)

- Aggregate operating balance: $\Pi_{\text{total}} = \sum_{f=1}^F m_f$.
- Aggregate household liquidity: $m_{\text{total}}^{HH} = \sum_{h=1}^H m_h$.
- Distribute dividend/deficit shares proportional to cash holdings:

$$\Delta m_h = \Pi_{\text{total}} \cdot \left(\frac{m_h}{m_{\text{total}}^{HH}} \right), \quad m_h \rightarrow m_h + \Delta m_h$$

- **Strict Firm Cash Reset:** Reset every firm cash balance identically to zero:

$$m_f \rightarrow 0.0 \quad (\forall f \in \{1, \dots, F\})$$

Step 3.3: Household Reservation Wage Update (PROC-3.3)

- **Employed Worker:** If wage earned $w_{\text{rec}} > \omega_h$, ratchet upward: $\omega_h \rightarrow w_{\text{rec}}$. If $w_{\text{rec}} \leq \omega_h$, keep ω_h unchanged (worker becomes unsatisfied; searches with probability 1.0 next month).
- **Unemployed Worker:** Apply statutory 10% markdown: $\omega_h \rightarrow 0.90 \cdot \omega_h$.

Step 3.4: Administrative Carryover & Deferred Firings (PROC-3.4)

1. **Execute Firings:** For each worker $h \in \text{pending_firings}_f$, sever Type B link (`employer_id = null`, remove from firm's employees). Clear `pending_firings = []`.
2. **Staffing Continuity:** If vacancies > 0 , set `unfilled_vacancy = true`, `consecutive_full = 0`. Else if vacancies $= 0$ and $l_f > 0$, set `unfilled_vacancy = false`, `consecutive_full += 1`. Reset vacancies $= 0$.
3. **Demand Rollover:** Copy $d_f^{\text{old}} \rightarrow \text{demand_current_month}$, then reset `demand_current_month = 0`.
4. **Clear Stockout Logs:** Empty `constrained_firms = {}` for all households.

5. Initialization State & Simulation Invariants

Defensive Programming & Verification Checks

5.1 Initialization Checklist (Equilibrium Start)

- To eliminate artificial simulation startup transients, initialize the state close to stationary levels:
- Total Money Supply:** Set $M_{\text{total}} = 100,000$. Allocate $m_h = \frac{M_{\text{total}}}{H} = 100.0$ to each household; set all firm cash $m_f = 0.0$.
 - Initial Employment:** Assign exactly 10 workers to each of the 100 firms ($\frac{1000}{100} = 10$). Initial unemployment is 0%.
 - Initial Wages & Reservations:** Set $w_f = 1.0$ for all firms; set $\omega_h = 1.0$ for all households.
 - Marginal Cost & Prices:** Marginal cost is $mc_f = \frac{1.0}{21.3} \approx 0.01587$. Set initial prices to a 10% markup: $p_f = 1.10 \cdot mc_f \approx 0.01746$.
 - Initial Inventories & Demand History:** Pre-fill warehouses with 1 month of normal production:

$$i_{f,0} = 21 \cdot \lambda \cdot l_f = 21 \cdot 3 \cdot 10 = 630 \text{ units}, \quad d_{f,0}^{\text{old}} = 630 \text{ units}$$

- Initial Network Graphs:** Draw $n = 7$ distinct supplier IDs uniformly at random for each household.

5.2 Strict Simulation Invariant Assertions

Embed these 6 mathematical invariant checks directly into your simulation engine loop:

✓ Simulation Invariant Assertions

1. Global Money Conservation:
 $|\sum_{h=1}^H m_h + \sum_{f=1}^F m_f - M_{\text{total}}| < 10^{-7}$

Verify at every daily and monthly tick t . Guarantees zero cash leakage during transactions, payroll, or dividend distribution.

2. Firm Cash Solvency:
 $m_f == 0.0 \quad \forall f \in \{1, \dots, F\}$

Verify on Day 1 morning of each month. All operating cash must be strictly cleared via PROC-3.2.

3. Household Budget Solvency:
 $m_h \geq 0.0 \quad \forall h \in \{1, \dots, H\}$

Verify before each purchase. Overdrafts strictly prohibited; trade bounded by cash-in-advance ($q^* \leq \frac{m_h}{p_f}$).

4. Non-Negative Inventory:
 $i_f \geq 0.0 \quad \forall f \in \{1, \dots, F\}$

Verify after daily trades. Stockout enforcement prevents warehouse inventory from dipping below zero.

5. Labor Accounting Invariant:
 $\sum_{f=1}^F l_f + |\{h : \text{employer}_h == \text{null}\}| == H = 1000$

Verify after matching (PROC-1.5) and firings (PROC-3.4). No agent may vanish from labor register.

6. Network Degree Preservation:
 $|\text{supplier_ids}_h| == 7 \quad \forall h \in \{1, \dots, H\}$

Verify after rewiring (PROC-1.4). Commercial network strictly preserves regular out-degree $n = 7$.

6. Implementation Verification & Stylized Facts

Model Validation Signatures (Lengnick 2013, Sec 3–5)

When your simulation is bug-free and executing properly, aggregate macroeconomic time-series will spontaneously reproduce the empirical stylized facts documented by Lengnick:

Empirical Stylized Fact	Expected Quantitative Signature in Model	Paper Citation
Spontaneous Coordination	Aggregate employment rate hovers near full employment: 95.8%...100% (unemployment $u \in [0\%, 4.2\%]$), demonstrating decentralized out-of-equilibrium stability.	Sec 3, Fig 6
Excess Demand Rarity	Unsatisfied consumption demand is $< 0.65\%$ in over 95% of all simulated months; shortages remain highly localized without systemic cascading failure.	Sec 3, Fig 5
Beveridge Curve	Robust negative empirical correlation between vacancy rate and unemployment rate (u vs v trade-off).	Sec 3, Fig 7b
Wage Phillips Curve	Negative correlation between unemployment rate and nominal wage growth / goods price inflation.	Sec 3, Fig 7a
Right-Skewed Firm Sizes	Firm workforce distribution is heavily right-skewed with sample skewness ≈ 1.72 and excess kurtosis ≈ 3.73 , matching empirical Zipf-type firm size distributions.	Sec 3, Fig 8
Calvo Price Stickiness	Median frequency of price changes across firms is $\approx 11\%$ per month, matching empirical microeconomic studies of sticky consumer prices.	Sec 3
GDP-Inflation Lag Dynamics	Real GDP positively correlates with inflation with a 2-quarter lag (peaking around 6 months), consistent with empirical VAR macro literature (Walsh 2003).	Sec 3, Fig 9
Long-Run Money Neutrality	Doubling nominal money stock M_{total} leads to exactly doubled price levels along a 45° ray with zero long-run effect on aggregate real output or employment.	Sec 4, Fig 11
Short-Run Money Non-Neutrality	A 5% positive money shock creates an immediate cyclical boom in real GDP peaking in year 1, decaying smoothly over 2 to 3 years before returning to baseline.	Sec 4, Fig 12
Endogenous Butterfly Effect	Perturbing just 1 household's demand by 5% on a single day causes the trajectory of aggregate GDP to fully diverge from baseline after several decades.	Sec 5, Fig 13